

Chapter 2: Mechanics of Futures Markets

1. Background and Operation (Section 2.1)

a. Forward Contracts vs. Futures Contracts

Feature	Futures Contract	Forward Contract
Trading Location	Traded on a Derivatives Exchange .	Traded in the Over-the-Counter (OTC) Market .
Contract Terms	Standardized by the exchange.	Can be customized if necessary.
Agreement	Obligation to buy/sell an asset at a future time for a certain price (both).	Obligation to buy/sell an asset at a future time for a certain price (both).

b. Major Futures Exchanges

- **CME Group:** Merger of Chicago Board of Trade (CBOT), Chicago Mercantile Exchange (CME), and New York Mercantile Exchange (NYMEX).
- **Other Large Exchanges:** InterContinental Exchange (ICE), Eurex, BM&F BOVESPA, and Tokyo Financial Exchange (TFX).

c. Creation of a Futures Contract Position

- **Long Position:** A trader (e.g., in New York) calls a broker with instructions to **buy** an asset for future delivery (e.g., 5,000 bushels of corn for September delivery).
- **Short Position:** Another trader (e.g., in Kansas) instructs a broker to **sell** the same amount of the asset for the same delivery month.
- **Price Determination:** A price is determined by the balance of **supply and demand** (the current futures price, e.g., 600 cents/bushel).
 - If more traders want to sell than buy, the price falls, attracting new buyers.
 - If more traders want to buy than sell, the price rises, attracting new sellers.
- **Trading Systems:**
 - **Open Outcry:** Traditional method where floor traders physically meet to determine the price.
 - **Electronic Trading:** A computer matches buyers and sellers.

d. Closing Out Positions

- **Vast Majority:** The **vast majority** of futures contracts do not result in delivery.

- **Definition:** Closing out a position means entering into the **opposite trade** to the original one, prior to the delivery period.
 - **Long Trader:** Closes out by **selling** (shorting) the same futures contract.
 - **Short Trader:** Closes out by **buying** (longing) the same futures contract.
- **Gain/Loss:** The trader's total gain or loss is determined by the **change in the futures price** between the initial trade date and the close-out date.
- **Importance of Delivery:** Although rare, the **possibility of final delivery** is what links the futures price to the spot price.

2. Specification of a Futures Contract (Section 2.2)

When an exchange develops a new contract, it must meticulously specify the exact nature of the agreement.

a. The Asset (Grade and Quality)

- **Commodities:** The exchange must stipulate the **grade or grades** of the commodity that are acceptable due to market variations in quality.
 - *Example (Orange Juice):* ICE specifies frozen concentrates that are US Grade A with a Brix value ≥ 62.5 degrees.
- **Adjustments:** For some commodities, a range of grades can be delivered, with the **price adjusted** based on the grade chosen.
 - *Example (Corn):* The standard is "No. 2 Yellow." Delivering "No. 1 Yellow" increases the price received by 1.5 cents/bushel.
- **Financial Assets:** Generally well-defined (e.g., Japanese yen).
 - *Exception (Treasury Bonds/Notes):* The contract allows delivery of any qualifying bond/note within a maturity range, with the price adjusted via a formula based on the coupon and maturity (discussed in Chapter 6).

b. The Contract Size

- **Definition:** Specifies the **exact amount of the asset** to be delivered under one contract.
- **Exchange Decision:** The size is a key decision:
 - **Too Large:** Excludes small investors/hedgers.
 - **Too Small:** Makes trading expensive due to per-contract costs.
- *Examples:*
 - Agricultural products: typically \$10,000 to \$20,000 in value.
 - Financial futures (Treasury bonds): face value of \$100,000.
- **Mini Contracts:** Exchanges introduce "mini" contracts (e.g., Mini Nasdaq 100) to attract **smaller investors** (retail traders).

c. Delivery Arrangements (Location)

- **Specification:** The exchange must specify the **place** where delivery will be made, especially for commodities with high transportation costs.
 - *Example:* ICE Orange Juice delivery is to exchange-licensed warehouses in Florida, New Jersey, or Delaware.
- **Short Position's Choice:** If alternative locations are specified, the **party with the short position** usually chooses the location.
- **Price Adjustment:** The price received by the short position may be adjusted (usually higher) for locations farther from the main source of the commodity.

d. Delivery Months and Trading Cessation

- **Reference:** A futures contract is referred to by its **delivery month**.
- **Specification:** The exchange specifies the exact delivery period (sometimes the whole month).
- **Choice:** Months are chosen to meet the needs of market participants (e.g., corn futures trade in March, May, July, September, and December).
- **Trading Cessation:** Trading generally ceases a few days **before** the last day delivery can be made.

e. Price Quotes

- The exchange defines how prices will be quoted (e.g., dollars and cents for crude oil, dollars and thirty-seconds of a dollar for Treasury bonds/notes).

f. Price Limits and Position Limits

- **Daily Price Limits:** Maximum allowed daily price movement specified by the exchange.
 - **Limit Down:** Price moves down by the daily limit from the previous close.
 - **Limit Up:** Price moves up by the daily limit from the previous close.
 - **Purpose:** To prevent large price movements due to speculative excesses. (Controversial, as they can become an artificial trading barrier.)
- **Position Limits:** Maximum number of contracts that a **speculator** may hold.
 - **Purpose:** To prevent speculators from exercising **undue influence** on the market.

g. Short Position's Choice (General Rule)

- When a contract specifies alternatives (e.g., grade or location), the **party with the short position** (the seller) generally chooses which option to exercise.
- The short party files a **notice of intention to deliver** with the exchange, specifying their selections.